

GOOGLE CLOUD RAPID AGENT HACKATHON · ELASTIC TRACK

IncidentIQ

The DORA major-incident triage **agent** — turn a 4-hour regulatory scramble into seconds, with a human always in control.

Gemini 3

Google Cloud Agent Builder

Elastic MCP

Vertex AI Agent Engine

Cloud Run

`incidentiq-908307939543.europe-west1.run.app`

THE PROBLEM

A regulator deadline starts the moment an incident hits

DORA requires a major-incident **early-warning report to the regulator within 4 hours**. Miss it, misclassify it, or file a sloppy one → **finances and supervisory scrutiny**.

Today it's **manual**: a compliance officer reads the incident, checks Article 18 thresholds, hunts for similar past incidents, decides MAJOR vs MINOR, and hand-writes the submission.

~2 hours · under deadline pressure · error-prone.

THE INSIGHT EVERYONE MISSES

Individually MINOR → MAJOR in aggregate

DORA has a rule most tools ignore: incidents that are individually MINOR become a **reportable major incident** when they **recur on the same service** within a reference window.

A string of "harmless" payment warnings is, legally, a major incident. Miss the pattern, miss the filing.

IncidentIQ encodes this rule. It's the difference between a demo and a compliance tool.

WHAT IT DOES — AN AGENT, NOT A CHATBOT

A multi-step plan that takes real actions

- **Search** similar incidents — hybrid kNN + keyword over Elasticsearch
- **Classify** against DORA Art.18 + the recurrence rule
- **Draft** the *actual* DNB submission (real template, not a summary)
- **Defend** — a version-stamped record a regulator can challenge
- **Wait** — consequential writes are **gated on human approval**
- **Act** — store classification, save obligations, write the audit trail

~Seconds vs ~2 hours.

THE "WOW" — LIVE IN THE DEMO

A routine warning becomes a regulatory MAJOR

On the dashboard, an innocuous card: "**DB warning: connection pool at 85% saturation.**" Zero client impact. Any tool calls it MINOR. Click it → IncidentIQ returns MAJOR.

"Would individually be MINOR... however classified MAJOR due to the DORA recurrence rule. This is the 6th PaymentProcessing incident in 30 days."

— generated live by the deployed Agent Builder agent

ARCHITECTURE · ALL THREE TECHS AT RUNTIME

Reason → plan → act → human gate → audit

```
Browser -POST /api/classify→ agent
  embed (Gemini) → precedent search (Elastic MCP `search`) → classify + recurrence
  (ES|QL)
  → explain + draft DNB (Agent Builder · Vertex AI Agent Engine) → defensibility +
  deadlines
  → PROPOSE (gated) —ApprovalBar→ writes + obligations + audit (Elastic)
```

Verifiable on the live [/health](#): `partner_mcp_connected:true · agent_builder_connected:true`

TECH STACK — REQUIRED TRIO, NO COMPETITORS

Gemini + Agent Builder + Elastic MCP

GEMINI 3

gemini-3-flash-preview

rationale, DNB draft, gemini-embedding-001 768-dim
vectors

ELASTIC MCP SERVER

@elastic/mcp-server-elasticsearch

spawned at runtime; hybrid kNN + keyword
precedent search

GOOGLE CLOUD AGENT BUILDER

Vertex AI Agent Engine

ADK agent, called via
reasoningEngines:streamQuery

HOST

Google Cloud Run

no competing AI or cloud services

WHY IT WINS THE ELASTIC TRACK

Elastic is load-bearing, not decorative

- Hybrid vector + keyword search **is** the precedent engine; ESIQL **STATS is** how the recurrence rule fires.
- Genuine agent behavior: multi-step plan → real tool actions → **human-in-the-loop** → audit trail.
- Real domain depth: DORA Art.18/19/28, the recurrence rule, a regulator-defensible record, the actual submission draft.
- **Verifiable runtime compliance** — all three required techs provably invoked live.

LIVE & OPEN

Seconds, not hours. Human always in control.

App (no login): incidentiq-908307939543.europe-west1.run.app

Repo (MIT): github.com/manojmallick/incidentiq